# VeriCBAM: Master Capstone Project Announcement

## Public LinkedIn Dissemination & Project Milestone Overview

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| **Degree Program:** | Master of Science in Data Science (M.Sc. DSc) | **Faculty Supervisor:** | Dr. Humera Noor \| Winter Semester 2026/2027 |
| **Live Cloud App:** | https://vericbam-uqpz4jrnpteqetfvjxtslh.streamlit.app/ | **Source Code:** | https://github.com/CaceresCristhian/VeriCBAM |

### LinkedIn Post Content (Approved Capstone Communication Milestone):

Excited to share my Master's Capstone project at the University of Europe for Applied Sciences (Potsdam Campus): VeriCBAM — A Data-Driven Decision Support System for Carbon Border Adjustment Mechanism (CBAM) Consistency Screening.

Starting in 2026, European industrial importers must verify self-reported greenhouse gas emissions for goods like Cement and Iron & Steel imported into the EU under Regulation (EU) 2023/956. However, commissioning physical on-site audits for thousands of overseas industrial installations poses severe logistical bottlenecks.

To address this challenge, VeriCBAM acts as an explainable screening copilot that cross-checks self-reported emissions declarations against independent multi-source evidence before costly inspections are commissioned:

- First-Principles Chemical Stoichiometry: Enforcing thermodynamic mass-balance lower bounds for cement clinker calcination and steelmaking reduction routes.
- Multi-Sensor Satellite Earth Observation: Integrating 30m USGS Landsat-8/9 thermal infrared (TIRS) core heat signatures, daily 375m NASA VIIRS active operating calendars, and high-resolution optical rail-yard freight activity to verify operational presence from orbit.
- Longitudinal Trajectories & Co-Pollutant Fingerprinting: Evaluating leave-one-year-out historical baselines and NOx/CO2 combustion ratios from public environmental compliance registries (EEA E-PRTR).
- Transparent Evidence Fusion: Synthesizing all streams through a confidence-gated Bayesian log-odds engine (achieving <3.5% false alarms on compliant facilities) and a supervised triage queue delivering a 2.71x audit prioritization lift.

The platform includes an interactive, browser-based decision cockpit with a 1-click Official Pre-Audit Screening Dossier (PDF) generator for customs declarants, accredited verification bodies, and industrial importers.

Explore the live interactive cloud cockpit: https://vericbam-uqpz4jrnpteqetfvjxtslh.streamlit.app/
Code & Documentation on GitHub: https://github.com/CaceresCristhian/VeriCBAM

Special thanks to my academic supervisor, Dr. Humera Noor, for her ongoing mentorship and guidance throughout this project!

**#DataScience #MachineLearning #RemoteSensing #CBAM #CarbonAccounting #EarthObservation #Sustainability #EUETS #AIforGood #UniversityOfEurope #PotsdamCampus**